

Recent Advanced Technologies in Financial Time-Series Generation: A Survey

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Abstract

The substantial capacity of large models (LMs) to process long sequences has sparked growing interest in time-series analysis, especially temporal modeling. In the financial domain, considerable efforts have been dedicated to generating time series, including product prices and booking orders over specific periods, significantly enhancing investors' decision-making capability. In this paper, we conduct a systematic review of advanced technologies for financial time-series generation (FTSG), aiming to provide a cutting-edge snapshot of this rapidly evolving field. Our main contribution is a hierarchical taxonomy. On the task horizon, we categorize the FTSG into three primary tasks distinguished by generation targets, including extrapolation, imputation, and data synthesis. For each task, we summarize the major problems that existing literature focuses on. Orthogonally, we introduce recently proposed works alongside techniques from traditional statistical and deep learning to contemporary approaches derived from generative AI, accompanied by comprehensive analyses. Additionally, we compile a collection of public high-quality datasets for FTSG, categorized by manufactural grains, from raw data sources to curated databases and benchmark datasets, and detail their key attributes. Finally, we discuss challenges and promising directions for each type of FTSG tasks. Our project is maintained on GitHub¹ to make persistent contributions to the community.

1 Introduction

Time-series analysis (TSA) [Zhang *et al.*, 2020; Islam *et al.*, 2024] is one of the central topics in the financial field, which mainly focuses on the modeling of market factors, including prices, returns, and trading volumes. TSA equips traders with comprehensive insights to enhance their understanding of marketing dynamics and risk perceptions. More importantly, a vital part of TSA - financial time-series data

generation (FTSG) [Liu *et al.*, 2025; Tanaka *et al.*, 2025; Wiese *et al.*, 2019], creates invisible time-series data derived from observed series, dedicated to trend forecasting and effectively guiding traders for rational decision-making.

The evolution of FTSG has mirrored the broader trajectory of artificial intelligence. Earlier approaches relied heavily on statistical probabilistic models, such as Auto-Regressive Integrated Moving Average [Qin *et al.*, 2025] and Generalized Autoregressive Conditional Heteroskedasticity [Kim *et al.*, 2021]. While these methods offer rigorous mathematical foundations and interpretability, they often struggle to capture the high-dimensional non-linearity and "stylized facts" of financial markets, which are widely known as heavy tails, leverage effect, and volatility clustering [Kwon and Lee, 2024]. The subsequent era of deep learning introduced deep neural networks (DNNs), especially Transformers, which significantly improved the ability to model temporal dependencies. However, DNN models are excellent in point-wise accuracy, while often failing to replicate the diverse range of potential market scenarios essential for robust risk management. The current landscape is shifting to Generative AI (GAI). Technologies such as Generative Adversarial Networks (GANs) [Naritomi and Adachi, 2020; Yoon *et al.*, 2019] and, more recently, Diffusion Models (DMs) [Ho *et al.*, 2020; Cao *et al.*, 2024], have demonstrated a remarkable capacity to generate high-fidelity financial data that remains indistinguishable from real market observations. Additionally, the adaptation of Large Language Models (LLMs) to the temporal domain births time-series foundation models (FMs) [Jin *et al.*, 2024; Liu *et al.*, 2025; Zhu *et al.*, 2025], enabling zero-shot generation and cross-domain knowledge transfer. However, despite these breakthroughs, financial data features a low signal-to-noise ratio, extreme non-stationarity, and "regime shifts" where market dynamics change abruptly due to geopolitical or macroeconomic events - significantly challenging recent techniques.

Existing surveys relevant to FTSG either focused on time-series forecasting [Kim *et al.*, 2025b; Kong *et al.*, 2025; Arsenault *et al.*, 2025], narrowed their scope to a specific technical paradigm [Nie *et al.*, 2024; Zhang *et al.*, 2024c], or catalogued synthetic financial data by data type and evaluation criterion under a privacy lens [Meldrum *et al.*, 2025]. Consequently, there remains a lack of a systematic review that treats the generative problems of financial time series

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¹<https://github.com/Lafitte1573/Awesome-FinTS-Generation>

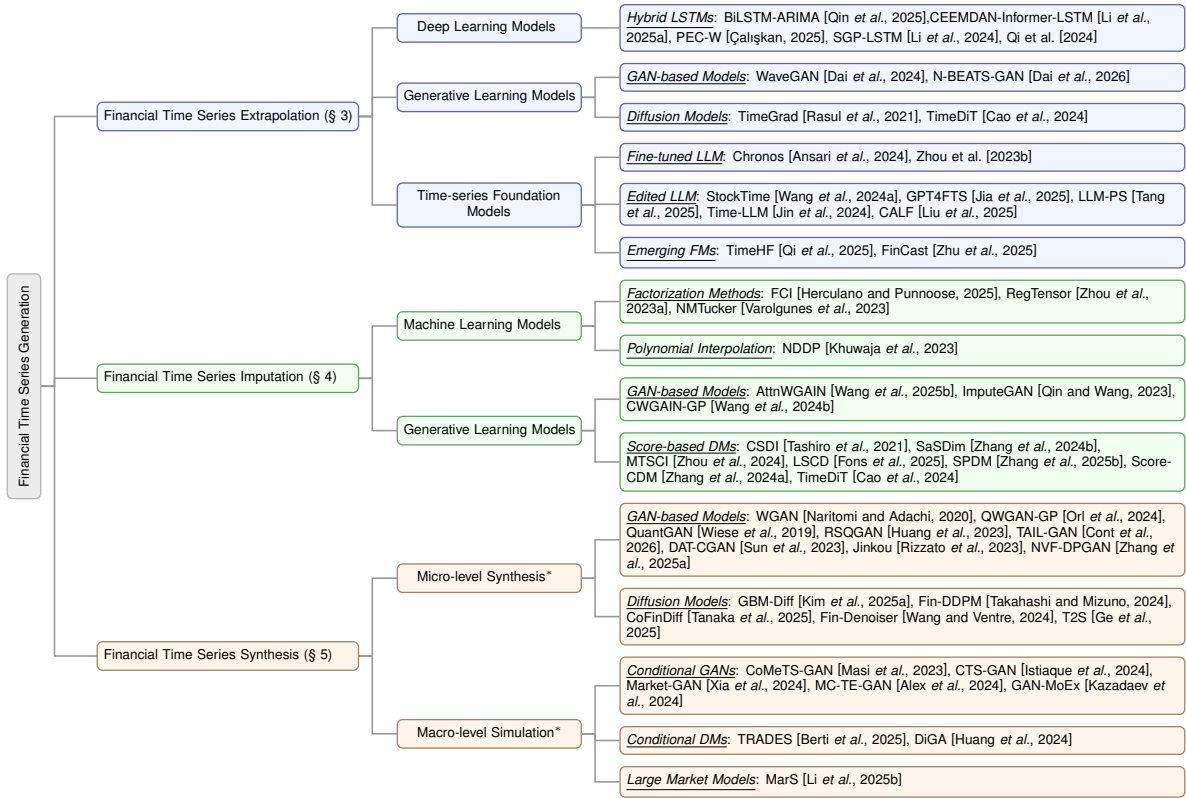


Figure 1: The overview of our taxonomy for FTSG. *: only generative learning models are mentioned.

as a task in their own right, organised through the lens of modern GAI. To bridge this gap, this paper systematically reviews literature published within *the last three years* to provide a cutting-edge snapshot of the FTSG landscape. Our contribution encompasses a two-level taxonomy that decouples FTSG into orthogonal *task* and *technique* horizons. At the task level, we categorize FTSG into three fine-grained types based on the data context: financial time-series extrapolation (FTSE) for extending short sequences, interval imputation (FTSI) for completing fragmentary data, and data synthesis (FTSS) for scaling data volume. At the technical level, we traverse a broad spectrum of methodologies, ranging from foundational statistics and machine learning to state-of-the-art GAI advancements, and showcase their shifting applications across FTSG tasks. By intersecting these two dimensions, we provide a structured analysis of how specific generative techniques are tailored to solve diverse financial data challenges. Furthermore, to facilitate community-wide adoption, we compile and categorize public resources - including data sources, curated databases, and benchmark datasets - based on their "degree of manufacturing." Finally, we summarize current challenges encountered by FTSG to outline promising directions for future research. The remainder of this paper is structured as follows: Section 2 formalizes the FTSG problem and provides an overview of our two-level taxonomy; Sections 3 to Section 5 offer a deep dive into the methodologies for FTSE, FTSI, and FTSS, respectively; Section 6 introduces our orchestration of high-quality data re-

sources; Section 7 identifies open challenges and future research frontiers; and Section 8 concludes the paper.

2 Taxonomy

We categorize the landscape of financial time-series generation into three primary tasks: extrapolation, imputation, and data augmentation. To facilitate illustration, we let $\mathbf{X} = \{x_1, x_2, \dots, x_T\}$ denote a multivariate financial time series in the below text, where $x_t \in \mathbb{R}^d$. Table 1 provides a systematic comparison of these tasks.

2.1 Time-Series Extrapolation

FTSE is a unidirectional, conditional generation process analogous to causal language modeling (CLM). Given an observed prefix $\mathbf{x}_{1:t}$, the goal is to model the conditional distribution of the subsequent sequence $\mathbf{x}_{t+1:T}$. In the financial domain, FTSE faces two unique hurdles: (1) the low signal-to-noise ratio and non-stationarity of market data, which contrasts with the structured logical flow of natural language, and (2) the inter-variable dependencies in high-dimensional series (e.g., cross-stock correlations), which add a layer of complexity absent in single-track token sequences.

Core Topics: FTSE is primarily utilized for generating future trajectories of financial time series, including stock prices, returns, interest rates, etc.

Table 1: Overview of the three recognized FTSG tasks.

Task	NLP Analogy	Sourced	Direction	Probabilistic Objective	Key Constraints	Main Target
FTSE	CLM	Yes	Unidirectional	$p(\mathbf{x}_{t+1:T} \mathbf{x}_{1:t})$	Non-stationarity	Price & Volatility Forecast
FTSI	MLM	Yes	Bidirectional	$p(\mathbf{x}_{miss} \mathbf{x}_{obs})$	Temporal Coherence	Data Repair & Cleaning
FTSS	-	No	Undirected	$p(\mathbf{X} \mathbf{z})$	Stylized Facts	Stress Testing & Simulation

2.2 Time-Series Imputation

In contrast to extrapolation, FTSI is a closed-format generation task analogous to masked language modeling (MLM). It aims to reconstruct missing or corrupted segments $\mathbf{x}_{miss}$ within a bounded sequence, leveraging bidirectional context from both preceding and succeeding observations. Unlike simple linear interpolation, FTSI must preserve the underlying stylized facts - such as volatility clustering - to ensure that the imputed segments are statistically indistinguishable from the surrounding observed data.

Core Topics: FTSI is mainly employed to repair incomplete time series data and perform "data cleaning" by replacing detected outliers with statistically plausible values.

2.3 Time-Series Synthesis

FTSS focuses on generating entirely synthetic sequences that mimic the underlying distribution $P_{data}(\mathbf{X})$ of real-world financial markets. Unlike FTSE or FTSI, FTSS is often a source-free generation process, frequently mapping a latent noise variable $\mathbf{z}$ from a prior distribution (e.g., Gaussian) to the data manifold: $\mathbf{X}_{syn} = G_{\theta}(\mathbf{z})$, $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$. This task is currently the fastest-evolving branch of FTSG, driven by GAI's ability to rebalance datasets or simulate rare "Black Swan" events that are underrepresented in historical data.

Core Topics: The applications of FTSS consist of macro-level market simulation for stress testing and micro-level synthetic data generation for training robust trading agents.

3 Financial Time Series Extrapolation

3.1 Deep Learning Models

Recurrent Neural Networks (RNN), particularly hybrid LSTMs [Qin *et al.*, 2025; Li *et al.*, 2025a], are widely used for FTSE by modeling the probability of future values based on historical observations. This essentially involves mapping time series into a latent space that encodes temporal dynamics, typically optimized via a Mean Squared Error (MSE) loss: $\mathcal{L}_{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$. However, since MSE tends to overfit transient fluctuations rather than persistent market regimes, vanilla RNNs often struggle with high-noise financial data. To address this, PEC-W [Çalışkan, 2025] integrates the Discrete Wavelet Transform (DWT) to decompose raw series into multi-resolution frequency components, which preserves structural temporal landmarks while filtering out high-frequency noises. Similarly, SGP-LSTM [Li *et al.*, 2024; Qi and others, 2024] employs Symbolic Genetic Programming to evolve high-quality features via a fitness function that balances returns, monotonicity, and the Information Coefficient (IC): $Fitness = TopR + \lambda_1 \times Monotonicity + \lambda_2 \times IC$. Following this, a filtering system selects stable "alpha" formulas (e.g., $ts_corr(close, volume, 10)$) which are then

fed into the LSTM. Apart from denoising techniques, Invariant Risk Minimization (IRM) [Cao *et al.*, 2023] is utilized to enhance robustness against distributional shifts. It augments the standard MSE loss with a relaxed IRM regulation loss $\gamma \|(\mathbf{w} \circ \nabla_{\mathbf{w}} R^e(\mathbf{w}, f))^{oc}\|_2^2$, which penalizes solutions that are not invariant across different market environments (e.g., bull vs. bear regimes).

Generative Learning Models. GAN-based models and diffusion models shift the focus from deterministic point-level forecasting to learning the global data distribution $p(\mathbf{x})$. WaveGAN [Dai *et al.*, 2024] combines Discrete Wavelet Transform (DWT) for denoising with Wasserstein GAN and Gradient Penalty (WGAN-GP) [Wang *et al.*, 2024b]. The optimization follows a minimax objective with a gradient penalty term:

$$\mathbb{E}_{\tilde{\mathbf{x}} \sim \mathbb{P}_g}[D(\tilde{\mathbf{x}})] - \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_r}[D(\mathbf{x})] + \lambda \mathbb{E}_{\tilde{\mathbf{x}} \sim \mathbb{P}_{\tilde{\mathbf{x}}}} \left[(\|\nabla_{\tilde{\mathbf{x}}} D(\tilde{\mathbf{x}})\|_2 - 1)^2 \right]$$

where λ penalizes deviations from the 1-Lipschitz constraint. N-BEATS-GAN [Dai *et al.*, 2026] integrates the interpretable basis expansion of N-BEATS [Oreshkin *et al.*, 2020] into a GAN generator to address multi-series, multi-output forecasting. By combining the GAN loss with the Transformation Invariant Loss with Distance Equilibrium, it effectively mitigates amplitude and phase distortions. Parallel to these, TimeGrad [Rasul *et al.*, 2021] introduced Denoising Diffusion Probabilistic Modeling to multivariate FTSE. By leveraging an RNN to condition a score-based decoder, it samples from the predictive distribution $p(\mathbf{x}_t | \mathbf{x}_{1:t-1})$ via Langevin dynamics. Following this idea, TimeDiT [Cao *et al.*, 2024] replaces the recurrent backbone with a Diffusion Transformer, which conditions on the observed time series ($\mathbf{x}^{con}$) to predict the future values ($\mathbf{x}^{tar}$) by μ -parameterization:

$$\mathcal{L} = \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t^{tar} | \mathbf{x}_0^{con})} \|\mu(\mathbf{x}_t^{tar}, \mathbf{x}_0^{con}) - \mu_{\theta}(\mathbf{x}_t^{tar}, t | \mathbf{x}_0^{con})\|^2$$

Notably, $\mathbf{x}_0^{con}$ and $\mathbf{x}_t^{tar}$ can indicate observed and corrupted values to support FTSI (§4).

3.2 Time Series Foundation Models

The latest studies borrow the idea of "pre-train then prompt" from NLP to FTSE. Instead of modeling probabilistic densities, researchers feed the price sequence to a foundational time series model (FM) and decode future values from the generated tokens. Given that most FMs are developed upon existing LLMs, substantial efforts are dedicated to bridging the gap between continuous real-valued time series and the discrete token space required by LLMs.

Fine-tuning is the most straightforward strategy to adapt LLM to time-series forecasting. As illustrated in Figure 2,

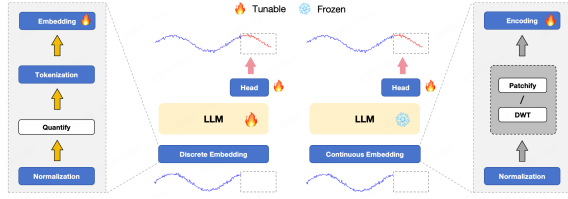


Figure 2: LLM fine-tuning (left) and editing (right) pipelines.

fine-tuning approaches preserve the core transformer architecture while updating pre-trained parameters to fit time series data. To align with the backbone LLM, Chronos [Ansari *et al.*, 2024] scales and tokenizes real values into a fixed vocabulary using quantile binning, and forecasting is treated as a cross-entropy classification task over discrete bins. In contrast, Zhou *et al.* [Zhou *et al.*, 2023b] give up discrete tokenization and divide time series into sub-series patches, which are then mapped into the LLM’s embedding space through a simple linear projection.

Model Editing approaches augment backbone LLMs with specialized modules to bridge the gap between numerical sequences and textual reasoning. To refine input representations, StockTime [Wang *et al.*, 2024a] maps 32-tick segments into the token space alongside hand-crafted textual summaries, while GPT4FTS [Jia *et al.*, 2025] utilizes an offline Dynamic Time Warping-based clustering to identify variable-length pattern boundaries, supported by an internal learnable wavelet convolution. To facilitate LLM understanding, LLM-PS [Tang *et al.*, 2025] introduces a *time-to-text* module to translate continuous intervals into semantic descriptions. To enable text-guided forecasting, Time-LLM [Jin *et al.*, 2024] reprograms frozen LLMs by aligning temporal patches with “text prototypes” via *Prompt-as-Prefix*. CALF [Liu *et al.*, 2025] employs a dual-branch architecture with consistency losses to minimize the distribution discrepancy between textual and temporal feature spaces.

Emerging Models. Unlike the mentioned LLM derivatives, some foundation models (FMs) are architected entirely from scratch for forecasting tasks. TimeHF [Qi *et al.*, 2025] leverages patch convolutional embeddings for long-term context and introduces a time-series policy optimization (TSPO) framework to incorporate human feedback for zero-shot generalization. FinCast [Zhu *et al.*, 2025] employs a sparse Mixture-of-Experts (MoE) backbone with learnable frequency embeddings. To mitigate market non-stationarity, it also utilizes a joint *Point-Quantile Loss*, enabling simultaneous optimization of deterministic accuracy and probabilistic risk for robust zero-shot forecasting across diverse assets.

4 Financial Time Series Imputation

4.1 Machine Learning Methods

Factorization Methods reconstruct missing data through low-rank approximation: the underlining principle is that a high-dimensional, partially observed matrix $\mathbf{X} \in \mathbb{R}^{m \times n}$ by two low-rank matrices $\mathbf{W} \in \mathbb{R}^{m \times r}$ and $\mathbf{H} \in \mathbb{R}^{r \times n}$ (where $r \ll \min(m, n)$), such that a missing entry $x_{i,j}$ is predicted by the inner product of corresponding factor vectors:

$\hat{x}_{i,j} = \mathbf{w}_i^\top \mathbf{h}_j$. FCI [Herculano and Punnoose, 2025] utilizes a Bayesian linear factor model $\mathbf{X} \approx \mathbf{\Lambda} \mathbf{F} + \mathbf{E}$, where Automatic Relevance Determination (ARD) priors are applied to the loading matrix $\mathbf{\Lambda}$ to automatically prune irrelevant components and prevent overfitting. To capture high-dimensional non-linearities, tensor decomposition is also introduced: $\mathcal{X} \approx \mathcal{G} \times_1 \mathbf{U}^{(1)} \dots \times_N \mathbf{U}^{(N)}$. For example, RegTensor [Zhou *et al.*, 2023a] integrates a Multi-Layer Perceptron (MLP) for non-linear factor interactions and employs Orthogonal Regularization to ensure distinct feature embeddings. In contrast, NMTucker [Varolgunes *et al.*, 2023] introduces a recursive “Matryoshka” Tucker decomposition, where the core tensor $\mathcal{G}$ is repeatedly decomposed across stacked layers with non-linear activations to improve parameter efficiency. In both paradigms, missing entries are filled using the corresponding indices in the reconstructed matrix $\hat{\mathbf{X}}$ or tensor $\hat{\mathcal{X}}$.

Polynomial Interpolation, such as the modified Newton-Divided Difference Polynomial (NDDP) [Khuwaja *et al.*, 2023], reconstruct missing observations by fitting a polynomial that accounts for non-linear momentum through the divided difference formula:

$$f[x_0, x_1, \dots, x_k] = \frac{f[x_1, x_2, \dots, x_k] - f[x_0, x_1, \dots, x_{k-1}]}{x_k - x_0}$$

where x and $f(x)$ denote timestamps and values, respectively. By calculating these higher-order differences across neighboring points, NDDP captures market acceleration and volatility more effectively than linear methods, providing a continuous, statistically consistent signal for downstream models like LSTMs.

4.2 Generative Learning Models

GAN-based Models formulate imputation as an adversarial game where a generator G maps a masked sequence $\tilde{\mathbf{X}} = \mathbf{X} \odot \mathbf{M}$ and noise $\mathbf{Z}$ to a completed series $\hat{\mathbf{X}} = G(\tilde{\mathbf{X}}, \mathbf{Z})$, and the discriminator D distinguishes between observed and imputed values. Standard GANs suffer from weak modeling of long-range spatio-temporal across variables and present distributional incoherence during contiguous outages. For the first problem, AttnWGAIN [Wang *et al.*, 2025b] replaces standard convolution with diagonal masked attention blocks within a WGAN framework, enabling the model to jointly capture temporal and feature-wise correlations. Similarly, ImputeGAN [Qin and Wang, 2023] employs an Informer-based generator that utilizes probabilistic sparse attention to manage long-sequence dependencies while reducing computational complexity. To combat the “drift” often seen during time-contiguous missing, CWGAN-GP [Wang *et al.*, 2024b] utilizes a Conditional WGAN with Gradient Penalty, which enforces Lipschitz continuity and ensures distributional stability across long-range missing windows.

Score-based Diffusion Models feature a forward diffusion step that gradually adds Gaussian noise to the partially observed series $\mathbf{X}_0$, and a reverse denoising step that reconstructs the original values, which is typically achieved by optimizing a simplified denoising objective:

$$\mathcal{L}_{diff} = \mathbb{E}_{\mathbf{X}_0, \epsilon, t} [\|\epsilon - \epsilon_\theta(\mathbf{X}_t, t, c)\|^2] \quad (1)$$

where ϵ_θ is a neural network trained to predict the noise ϵ added at timestep t and the condition c incorporates the observed data. To improve structural consistency modeling, LSSDM [Liang *et al.*, 2025] conducts the denoising process in a lower-dimensional latent space to align imputed values with the data’s underlying manifold. To resolve the ”boundary disharmony” problem, where jumps occur between observed and imputed regions, specialized consistency constraints are integrated into the denoising process, utilizing dependencies from temporal [Tashiro *et al.*, 2021] and spatial horizon [Zhang *et al.*, 2024b] and cross variables [Zhou *et al.*, 2024] to ensure smooth, mathematically aligned with the observed context. Additionally, given that financial data often exhibits cyclical or seasonal patterns difficult to capture in the time domain, LSCD [Fons *et al.*, 2025] and SPDM [Zhang *et al.*, 2025b] utilize frequency-domain transformations to map 1D time series into 2D periodic representations, effectively isolating stable cyclical components from high-frequency noise. Score-CDM [Zhang *et al.*, 2024a] complements this by employing a score-weighted convolution module to adaptively capture global temporal features, alongside an adaptive reception module that dynamically tunes receptive fields within the spectral domain to balance local volatility with long-term seasonal trends.

5 Financial Time Series Synthesis

Financial time series synthesis (FTSS) is categorized into two main tasks: micro-level and macro-level. Micro-level FTSS aims to generate price sequences along the temporal axis, while macro-level FTSS simulates the evolving market by generating order flows. The detailed comparison of these two tasks is illustrated in Figure 3.

5.1 Micro-level Time-series Synthesis

Generative adversarial networks. The direct application of standard GANs often faces training instability and mode collapse due to the unique challenges of financial data, i.e., ”stylized facts” [Kwon and Lee, 2024; Safari *et al.*, 2025]. A crucial improvement is WGAN (WGAN) [Naritomi and Adachi, 2020], which replaces the standard Jensen-Shannon divergence loss of GAN with the Wasserstein-1 distance:

$$\begin{aligned} & \mathbb{E}_{x \sim \mathbb{P}_r} [\log D(x)] + \mathbb{E}_{\tilde{x} \sim \mathbb{P}_g} [\log(1 - D(\tilde{x}))] \\ \rightarrow & \inf_{\gamma \in \Pi(\mathbb{P}_r, \mathbb{P}_g)} \mathbb{E}_{(x, \tilde{x}) \sim \gamma} [\|x - \tilde{x}\|] \end{aligned}$$

where $\Pi(\mathbb{P}_r, \mathbb{P}_g)$ denotes the set of all joint distributions $\gamma(x, \tilde{x})$ whose marginals are $\mathbb{P}_r$ and $\mathbb{P}_g$, respectively. This modification enhances training stability and allows for a meaningful gradient even when the distributions are non-overlapping in high-dimensional financial data spaces. Following the success of WGAN, subsequent financial GANs refine architectures and loss functions to advance the modeling of stylized facts. Quant-GAN [Wiese *et al.*, 2019] replaces traditional RNN backbones with temporal convolutional networks (TCNs) to capture long-range dependencies and *volatility clustering*. TAIL-GAN [Cont *et al.*, 2026] leverages the joint elicibility of Value-at-Risk (VaR) and Expected Shortfall (ES) to address *tail-risk-sensitive* caused

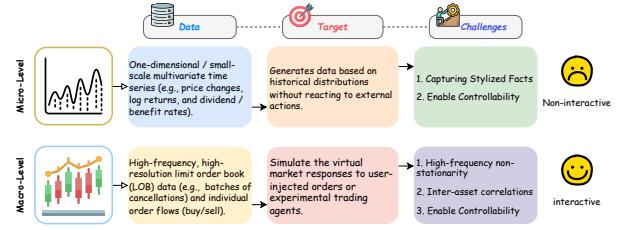


Figure 3: Comparison of micro and macro-level FTSS.

by extreme market events. On the other hand, market dynamics are considered in the generation. RSQGAN [Huang *et al.*, 2023] extends the Quant-GAN architecture by conditioning on market regimes (e.g., bull vs. bear), allowing the generator to produce high-fidelity ”deepfakes” that reflect specific historical or synthetic economic conditions. Jinkou [Rizzato *et al.*, 2023] introduces a similar conditional framework that replicates multivariate distributions by setting bounds on ”state variables” - global indicators common to all assets at a given timestamp.

Diffusion models are adapted to generate high-fidelity financial time series by incorporating domain-specific knowledge and data transformation techniques. The standard DMs assume additive, time-homogeneous Gaussian noise, which is incompatible with the dynamics of asset prices. GBM-Diff [Kim *et al.*, 2025a] integrates Geometric Brownian Motion (GBM), the core mechanism of the Black-Scholes framework, directly into the forward noising process to accurately replicate key stylized facts. Inspired by visual generation, Takahashi *et al.* [2024] and CoFinDiff [Tanaka *et al.*, 2025] leverage the exceptional 2-D image generation capabilities of DDPMs to model the complex, multivariate dependencies of financial data. Specifically, the Haar wavelet is used to decompose multivariate series into space-time representations, mapping different assets or indicators to distinct color channels. This transformation allows the DDPM to learn joint distributions and cross-correlations as spatial patterns, which are then reconstructed into realistic time series via the Inverse Discrete Wavelet Transform (IDWT). CoFinDiff further conditions the diffusion process on specified market attributes (e.g., trend or volatility levels). It employs classifier-free guidance to ensure the synthetic outputs strictly adhere to the given conditions while maintaining the ”stylized facts” of real-world financial data. The other conditional DMs for FTSS are: Fin-Denoiser [Wang and Ventre, 2024], which conditions on corrupted financial time series to generate synthetic data with a high signal-to-noise ratio; and T2S [Ge *et al.*, 2025], a diffusion transformer generating high-resolution time series controlled by natural language descriptions.

5.2 Macro-level Market Simulation

Market simulation aims to replicate the complicated statistics and temporal dynamics of real financial markets, thereby facilitating the validation of trading strategies and the assessment of risk. This is achieved by order-level generation and encounters challenges: modeling high-frequency non-stationarity, capturing subtle inter-asset correlations, and enabling controllability to simulate specific market conditions.

Conditional GAN is widely used in market simulation to enable controllability. The main difference among distinct GAN-based FTSS models is contingent on market conditions. CoMeTS-GAN [Masi *et al.*, 2023] generates multivariate time series (e.g., mid-prices and volumes for multiple correlated stocks) conditioned on a past sequence of observations. CTS-GAN [Istiaque *et al.*, 2024] generates out-of-sample multivariate time series conditioned on market states. Market-GAN [Xia *et al.*, 2024] focuses on generation controlled by a broader semantic context, including market dynamics or specific stock identifiers. It employs a two-stage training scheme with an autoencoder to learn low-dimensional features and several supervisors for knowledge transfer. MC-TE-GAN [Alex *et al.*, 2024] extends this conditioning by explicitly incorporating macroeconomic factors to simulate large-scale market scenarios like crashes or bull runs, focusing on concurrently generating multiple return series while preserving univariate properties (e.g., heteroskedasticity). Beyond conditions modeling, GAN-MoEx [Kazadaev *et al.*, 2024] incorporates temporal convolutional networks (TCNs) to capture temporal dependencies in simulating multidimensional log-returns.

Conditional Diffusion Model. TRADES [Berti *et al.*, 2025] is a conditional diffusion model (CDM) tailored for generating realistic and responsive limit order book (LOB) order flows in the high-frequency context. It employs a Transformer-based architecture to effectively capture the temporal and spatial characteristics of LOB data, training the model using the standard ϵ -parameterization loss function (Eq. 1). Additionally, DiGA [Huang *et al.*, 2024] introduces a two-stage controllable generation framework: it first utilizes a diffusion model to capture the evolution of market states (e.g., mid-price returns and order arrival rates) and then employs a meta-agent, guided by these latent states and financial economic priors, to synthesize high-fidelity order flows.

Large Market Model. Li *et al.* [2025b] proposed the first large market model (LMM), MarS, an order-level generative foundation model for market simulation. Unlike the mentioned generative models, MarS scales across massive order sequences to capture the asynchronous nature of market microstructure. Consequently, it provides a realistic, interactive, and controllable environment for evaluating trading behaviors with minimal financial risks.

6 Data Resource

Time series data in the financial domain exhibits high diversity and long temporal features due to the frequent daily transactions within numerous active markets around the world. According to the degree of post-processing, publicly available data material can be divided into raw *data sources*, curated *databases*, and benchmark *datasets*.

Data sources. Financial time series encompass diverse prices and economic indicators derived from real-world financial markets, which are typically categorized into four classes: 1) *Stocks & Index funds*. Index funds (e.g., S&P 500, NASDAQ 100, DOW Jones, and HS 300) and individual stocks (e.g., AAPL, TSLA, AMZN) are most commonly adopted in FTSG. These data exhibit significant

volatility clustering, *fat tails* in their return distributions, and *non-stationarity* in the raw price series, which necessitate a logarithmic transformation of returns in data processing. 2) *Macroeconomics*. Macro indicators, like treasury yields, inflation rates (CPI), gross domestic product (GDP), and fixed income, indicate the systemic economic health of society and generally have a relatively low frequency (daily, monthly, or quarterly) and are crucial for asset pricing models, risk management, and forecasting economic cycles. 3) *Global trades* represented by exchange rates and foreign exchange and commodities, encompassing raw materials and cross-currency risk. They are highly liquid and available in high-frequency formats, exhibiting complex autoregressive patterns. 4) *High-frequency trades*. Order books, tick-by-tick trades, and quotes are recorded at the millisecond or microsecond level and are essential for market microstructure analysis and the development of high-speed algorithmic trading strategies.

Databases. While raw financial time series are abundant, they are often fragmented across disparate platforms and optimized for human visualization rather than automated processing. Databases serve as essential intermediaries that bridge this gap by consolidating raw sources into structured, machine-readable formats accessible via robust APIs or programmatic portals. 1) *Financial Portals*: Platforms like Yahoo Finance³, Investing.com⁴, and Wind⁵ provide accessible historical market data across global equities and currencies. Their open-source wrappers, i.e., yfinance⁶ and investpy⁷, enable researchers to freely use these data. 2) *Developer-Centric APIs*: Alpha Vantage⁸ and Alpaca⁹ offer cloud-based delivery of real-time and historical data, often including technical indicators (e.g., SMA, RSI) and brokerage integration for automated trading. 3) *Institutional Repositories*: FRED¹⁰ (maintained by the Federal Reserve Bank of St. Louis) is the gold standard for U.S. macroeconomic data. For Chinese markets, CS-MAR¹¹ provides comprehensive, high-quality data for academic research on listed companies and A-share trading.

Datasets. Thanks to the encapsulation of databases, researchers can create task-specific datasets by customizing the sampling process of real-world time series for model training and evaluation. Furthermore, authoritative benchmarks complement this by pre-defining train/test splits and evaluation metrics. This standardization ensures a "level playing field," allowing for a fair and reproducible comparison of learning algorithms and model architectures. The most recent benchmarks, coupled with their key statistical characteristics, are detailed in Table 2.

³<https://finance.yahoo.com/>

⁴<https://www.investing.com/>

⁵<https://www.wind.com.cn/>

⁶<https://github.com/ranaroussi/yfinance>

⁷<https://github.com/alvarobartt/investpy>

⁸<https://www.alphavantage.co/>

⁹<https://alpaca.markets/>

¹⁰<https://fred.stlouisfed.org/>

¹¹<https://data.csmar.com/>

Table 2: Summary of recent FSTG benchmarks. †: the ratio of train: valid: test. ‡: the number of time series within a pre-defined time span.

Dataset	Task	Data Sources	Split [†]	Total [‡]
FinTSB [Hu <i>et al.</i> , 2025]	FTSE / FTSI	Chinese A-share, NYSE, NASDAQ	7:1:2	150M
GIFT-EVAL [Aksu <i>et al.</i> , 2024]	FTSE / FTSI	FRED, Exchanges, Bitcoin, NYSE, NASDAQ, NN5	9:0:1	25M
TFB [Qiu <i>et al.</i> , 2024]	FTSE / FTSI	FRED, Exchanges, NN5, NYSE, NASDAQ	7:1:2	8,068
M-Competitions ²	FTSE / FTSI	Walmart Sales	49:0:1	1,941
FinTSBridge [Wang <i>et al.</i> , 2025a]	FTSE / FTSI	S&P 500, NASDAQ, Dow Jones, FTSE, CSI 300 ETF, Bitcoin	7:1:2	86,978
TSGBench [Ang <i>et al.</i> , 2023]	FTSS	GOOGL, Exchanges	9:0:1	13,213
CTBench [Ang <i>et al.</i> , 2025]	FTSS	Cryptocurrencies on Binance Exchange	9:0:1	530
FinMultiTime [Xu <i>et al.</i> , 2025]	FTSE / FTSI	S&P 500, HS 300	-	6M

7 Challenges and Future Directions

We outline the common challenges encountered by FTSE, FTSI, and FTSS, and discuss potential improvements for future studies.

7.1 Causal and Invariant Representation Learning

Current FTSG models learn spurious correlations specific to a particular market environment or time period, because of the lack of explicit disentangling true causal factors (e.g., changes in fundamental economic policy) from mere coincidental patterns (e.g., a specific trading algorithm’s temporary dominance). Consequently, models that assume data are Independent and Identically Distributed (i.i.d.) often fail drastically when encountering a novel market regime (distributional shifts). Future FTSE frameworks may treat specific market regimes as separate training “environments” based on volatility, interest rate levels, or sentiment indices. Invariant Risk Minimization and Reinforcement Learning can be leveraged to retain the model’s core capacity only on the features that remain invariant across diverse environments.

7.2 Physics-informed Generative Modeling

A constraint of pure data-driven generative learning models (GANs and DMs) is that they can replicate past empirical distributions without guaranteeing adherence to established market microstructure or stochastic process theory, e.g., ensuring returns are scale-invariant or volatility is mean-reverting. Embedding priors - such as the Geometric Brownian Motion [Kim *et al.*, 2025a] or jump-diffusion processes for extreme events - into architectures provides a powerful white-box constraint. This shift enhances interpretability and ensures the generated sequences are not merely plausible but are also economically rational, which is crucial for regulatory stress-testing and reliable risk modeling.

7.3 Multi-modal Time Series Generation

Existing FTSG models broadly generate time-series data in isolation, often neglecting the exogenous signals that drive market dynamics. However, financial markets are inherently multi-causal systems: a macroeconomic announcement or a sudden news event (textual sentiment) triggers shifts in order flow (discrete events), which subsequently alter implied volatility surfaces (continuous manifolds). We identify two promising directions to capture this complexity:

Cross-modal Conditioned Generation. To create contextually grounded synthetic data, models must move beyond univariate history. While pioneering efforts like

TimeLLM [Jin *et al.*, 2024] and CALF [Liu *et al.*, 2025] utilize narratives of the market to guide generation, these approaches are often hampered by limited context windows and non-standardized narrative language. Future works can encode heterogeneous data, such as central bank operations or earnings call transcripts, using a unified lingual system or representation space that can precisely steer the generative trajectory of market factors.

Joint Cross-modal Generation. Except for treating non-TS data as static conditions, a promising approach involves the *simultaneous* generation of linked data streams. For example, by jointly modeling Level-2 limit order books, news headlines, and derivative prices, generators can inherently learn cross-modal causal dependencies and lead-lag relationships. This is particularly vital for simulating high-impact scenarios such as flash crashes or liquidity crises, where the feedback loop between information arrival and price discovery determines the market outcome.

7.4 Online Adaptation

Despite the zero-shot capabilities of modern time-series FMs, they remain susceptible to concept drift - the frequent distributional shifts triggered by evolving economic cycles or regulatory changes. Therefore, FTSG must transition toward online adaptation to avoid the notorious “catastrophic forgetting.” One promising avenue is Bayesian Continuous Learning, utilizing Streaming Variational Bayes to incrementally update the generator’s latent space as new market observations arrive. Alternatively, framing generation within an Online Reinforcement Learning framework, where the model is rewarded for the real-time fidelity of synthetic samples relative to current market volatility, allows for near-instant calibration to abrupt regime shifts.

8 Conclusion

This survey presents a systematic review of the financial time-series generation (FTSG) landscape. Our main contribution is an enlightening, two-level taxonomy that unifies the field across both task-oriented and technique-driven dimensions, providing a structured framework for future studies. Furthermore, we characterize the data resource by categorizing manufacturing degrees into raw sources, curated databases, and carefully crafted benchmarks. By outlining promising directions to the unique challenges, we hope this work offers critical insights and advances the research community.

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